PYTHON WITH AI
MINI PROJECT REPORT

STUDENT PERFORMANCE ANALYTICS SYSTEM

A Python-based system for analysing student marks, grades and attendance

Project Detail	Information
Student Name	Diksha Ahirwar
Course	Python with AI
Project Type	Mini Project
Language	Python 3
Libraries	Pandas and NumPy
Dataset	20 fictional sample student records

This single PDF contains the complete project report, source code, sample student dataset, project documentation and README information. GitHub upload instructions are intentionally not included.

DECLARATION

I hereby declare that this mini project report titled **“Student Performance Analytics System”** has been prepared for academic learning as part of the Python with AI course. It demonstrates the use of Python, Pandas and NumPy to analyse sample student marks and attendance. The dataset used in this project contains fictional records prepared for demonstration.

Student Name: Diksha Ahirwar

Date: __ 14 October 2026

ACKNOWLEDGEMENT

I would like to express my gratitude to my course instructor for providing the opportunity to work on this mini project. This project helped me practise Python programming, data handling, conditional logic, loops, functions and basic data analysis. I also acknowledge the official Python, Pandas and NumPy documentation as learning resources.

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1. ABSTRACT

The Student Performance Analytics System is a beginner-level data analysis project developed using Python 3, Pandas and NumPy. It reads student records from a CSV file and analyses marks in Python, Mathematics and Data Analysis, along with attendance percentage. The system calculates total and average marks, assigns grades, identifies pass/fail status, checks attendance against a defined threshold, generates class-level and subject-wise summaries, identifies the highest and lowest performers, and lists the top five students. The processed results can be exported to a new CSV report. This project demonstrates how basic programming concepts and data analysis libraries can convert raw records into a structured report.

Keywords

Python, Pandas, NumPy, CSV, student performance, data analysis, grades, attendance.

2. INTRODUCTION

Educational institutions maintain student information such as subject marks and attendance. Reviewing each record manually can take time and make it difficult to identify overall class trends. A simple automated analysis system can organise this information and produce consistent calculations.

This project uses Python to process a sample dataset of 20 fictional students. Pandas reads and manages tabular data, while NumPy performs numerical calculations. Python functions, loops and conditional statements implement the grading and result rules. The project is an educational demonstration and is not a replacement for an institution’s official examination system.

3. PROBLEM STATEMENT

Raw student records show individual marks and attendance, but do not automatically provide totals, averages, grades, pass/fail results or class summaries. Manually calculating these values for every student can be repetitive. This project addresses the problem by automating common calculations and generating a standardised performance report.

4. OBJECTIVES

- Read student information from a CSV file using Pandas.
- Calculate total and average marks across three subjects using NumPy.
- Assign a grade based on each student’s average marks.
- Determine pass/fail status using a minimum mark requirement in every subject.
- Classify attendance using a 75% threshold.
- Calculate class average and subject-wise averages.
- Identify the highest and lowest averages and display the top five students.
- Export processed records to a CSV report.

5. SCOPE AND SIGNIFICANCE

The project is designed for small, structured student datasets and for learning core programming concepts. It can help a user quickly review a sample class, compare subject averages and locate records that may need attention. Its scope is limited to descriptive analysis; it does not predict future performance or use machine learning.

Main outputs

- A student-wise table showing total, average, grade, result and attendance status.
- A class summary with student count, passes, failures and average marks.
- Subject-wise average marks in descending order.
- A top-five student list.
- A saved file named student_performance_report.csv.

6. TOOLS AND TECHNOLOGIES

Technology	Purpose
Python 3	Main programming language; functions, loops, conditions and error handling.
Pandas	Reads CSV data into a DataFrame and performs table-based analysis.
NumPy	Calculates total and average marks efficiently using arrays.
CSV	Stores the input student dataset and processed output report.
Python runtime	Runs the code in a compatible Python environment.

7. DATASET DESCRIPTION

The input file is named **students.csv** and contains 20 fictional student records. Each record includes an ID, name, department, marks in three subjects and attendance percentage.

Column	Description
Student_ID	Unique sample identifier for each student.
Name	Fictional student name.
Department	Department label: Arts, Commerce or Science.
Python	Marks in Python, on a 0–100 scale.
Mathematics	Marks in Mathematics, on a 0–100 scale.
Data_Analysis	Marks in Data Analysis, on a 0–100 scale.
Attendance_Percent	Attendance percentage, from 0 to 100.

Data note: Names and marks are fictional sample data created only to demonstrate the program. They are not actual student records.

8. SYSTEM DESIGN AND METHODOLOGY

Step	Operation	Description
1	Load data	Pandas reads students.csv into a DataFrame.
2	Calculate marks	NumPy calculates row-wise totals and averages for three subjects.
3	Assign grade	A reusable function assigns a grade from the average.
4	Check result	A student passes only if every subject mark is at least 40.
5	Check attendance	Attendance of 75% or more is marked Satisfactory.
6	Summarise	The program calculates class and subject statistics and identifies top/bottom performers.
7	Export	The completed DataFrame is saved as student_performance_report.csv.

Algorithm

- Start the program.
- Load students.csv into a Pandas DataFrame.
- Select the three subject columns and convert them to a NumPy array.
- Calculate each student’s total marks and average marks.
- Use assign_grade() to assign a grade.
- Loop through every student to determine pass/fail and attendance status.
- Display student-wise results, class statistics, subject averages and top five students.
- Save all processed records to student_performance_report.csv.
- Handle missing input file or required column/value errors with a clear message.
- End the program.

9. GRADING AND EVALUATION RULES

The following grading scale is used for this educational project. It may differ from a school, college or university's official rules.

Average Marks	Grade
90 and above	A+
80 to below 90	A
70 to below 80	B
60 to below 70	C
50 to below 60	D
Below 50	E

Pass/fail rule

A student is marked **Pass** only when their marks are at least 40 in each of the three subjects: Python, Mathematics and Data Analysis. If any one subject is below 40, the student is marked **Fail**.

Attendance rule

Attendance of 75% or above is marked **Satisfactory**. Attendance below 75% is marked **Below 75%**. Attendance status is reported separately and does not by itself determine the pass/fail result.

10. IMPLEMENTATION AND CODE EXPLANATION

Function / Component	Role
load_data(filename)	Reads the CSV file using pandas.read_csv().
assign_grade(average)	Uses if/elif/else conditions to return a grade.
process_data(data)	Uses NumPy to calculate totals and averages, then loops through records to assign results and attendance status.
show_analysis(data)	Prints the student report and summaries, then saves the output CSV.
main()	Coordinates the steps and handles common file/dataset errors.

Programming concepts used

- **Variables and constants:** DATA_FILE, SUBJECTS and PASS_MARK store reusable configuration values.
- **Data types:** strings, numbers, lists and DataFrames represent names, marks, subject columns and records.
- **Functions:** separate the program into reusable and readable units.
- **Loops:** iterate through students and subject averages.
- **Conditions:** apply grading, pass/fail and attendance rules.
- **NumPy arrays:** calculate row-wise total and mean marks.
- **Pandas DataFrame:** stores, filters, summarises and exports tabular data.
- **Exception handling:** displays a helpful message when the input file or required columns are unavailable.

11. RESULTS AND ANALYSIS

Measure	Result
Total students	20
Passed	18
Failed	2
Class average (average marks)	68.77
Highest student average	94.33
Lowest student average	38.67
Top performer	Ira Mishra (94.33)
Lowest average	Arjun Rao (38.67)

Subject-wise average marks

Subject	Average Marks
Data_Analysis	69.60
Mathematics	68.90
Python	67.80

Interpretation: In this sample dataset, Data Analysis has the highest subject-wise average (69.60), followed by Mathematics (68.90) and Python (67.80). The overall class average is 68.77. Eighteen students meet the project's pass rule and two do not. These findings describe only the fictional sample data.

Top five students

Rank	Student	Average	Grade
1	Ira Mishra	94.33	A+
2	Ananya Patel	91.67	A+
3	Rohan Yadav	89.33	A
4	Diya Verma	88.00	A
5	Krish Nair	84.67	A

Example student record

Ira Mishra scored 95 in Python, 96 in Mathematics and 92 in Data Analysis. The total is 283 out of 300 and the average is 94.33. The assigned grade is A+, the result is Pass, and attendance of 99% is marked Satisfactory.

When the program runs successfully, it saves the processed dataset as **student_performance_report.csv**. The exported file contains original student details plus total marks, average marks, grade, pass/fail result and attendance status.

12. TESTING

Test case	Expected behaviour	Program logic
Input file available	Read CSV and process records.	Reads students.csv.
All subject marks are 40 or higher	Result is Pass.	Checks all three subjects.
Any subject mark below 40	Result is Fail.	Fail if one subject is below 40.
Attendance >= 75%	Satisfactory attendance.	Threshold is >= 75.
Attendance < 75%	Below 75% status.	Threshold condition labels it.
Input file missing	Display helpful error.	FileNotFoundError is handled.
Required column missing/invalid value	Display dataset error.	KeyError and ValueError are handled.

13. CONCLUSION

The Student Performance Analytics System demonstrates how Python can automate common calculations on student marks and attendance. Pandas handles tabular data and NumPy performs numerical operations. The program produces student-wise totals and averages, grades, pass/fail results, attendance classifications, class statistics, subject averages and a top-five list. It also exports the processed results to a CSV file. This mini project provides practical experience with functions, loops, conditional statements, arrays, tabular data and basic error handling.

14. LIMITATIONS AND FUTURE SCOPE

Limitations

- The dataset is a small fictional sample of 20 students.
- The grading and pass rules are illustrative and may not match official institutional policies.
- The program analyses the three specified subjects only.
- The current system is command-line based and has no graphical user interface.
- The project describes performance but does not predict future results.

Possible future improvements

- Allow users to enter new student records through a form.
- Add charts for marks, attendance and subject comparisons.
- Validate that marks and attendance stay within the 0–100 range.
- Support additional subjects and configurable grading policies.
- Create a simple dashboard or web interface.
- Use larger, authorised datasets while protecting student privacy.

15. LEARNING OUTCOMES

- Practised reading CSV files and working with DataFrames.
- Used NumPy functions to calculate totals and averages.
- Applied functions, loops, variables and conditional statements.
- Learned to produce summaries and rank records by average marks.
- Exported processed data into a new CSV file.
- Understood how to document and test a small data-analysis workflow.

APPENDIX A: PYTHON CODE — main.py

This is the complete source code used for the project. To execute it, save the code as **main.py** and keep **students.csv** in the same folder. Pandas and NumPy must be installed in the Python environment.

```
"""
Student Performance Analytics System
Python with AI mini project
"""
import numpy as np
import pandas as pd

DATA_FILE = "students.csv"
SUBJECTS = ["Python", "Mathematics", "Data_Analysis"]
PASS_MARK = 40

def assign_grade(average):
    """Assign a grade using the student's average marks."""
    if average >= 90:
        return "A+"
    elif average >= 80:
        return "A"
    elif average >= 70:
        return "B"
    elif average >= 60:
        return "C"
    elif average >= 50:
        return "D"
    return "E"

def load_data(filename):
    """Read student records from a CSV file using Pandas."""
    return pd.read_csv(filename)

def process_data(data):
    """Calculate totals, averages, grades, results and attendance status."""
    marks = data[SUBJECTS].to_numpy()
    data["Total_Marks"] = np.sum(marks, axis=1)
    data["Average_Marks"] = np.round(np.mean(marks, axis=1), 2)

    grades = []
    results = []
    attendance_status = []

    for _, student in data.iterrows():
        grades.append(assign_grade(student["Average_Marks"]))

        if all(student[subject] >= PASS_MARK for subject in SUBJECTS):
            results.append("Pass")
        else:
            results.append("Fail")

        if student["Attendance_Percent"] >= 75:
            attendance_status.append("Satisfactory")
        else:
            attendance_status.append("Below 75%")

    data["Grade"] = grades
    data["Result"] = results
    data["Attendance_Status"] = attendance_status
    return data

def show_analysis(data):
```

APPENDIX A: PYTHON CODE — main.py (continued)

```
    """Display the report and save the processed dataset."""
    print("\nSTUDENT PERFORMANCE ANALYTICS SYSTEM")
    print("=" * 60)
    columns_to_show = [
        "Student_ID", "Name", "Department", "Total_Marks",
        "Average_Marks", "Grade", "Result", "Attendance_Percent"
    ]
    print(data[columns_to_show].to_string(index=False))

    print("\nCLASS SUMMARY")
    print("Total students:", len(data))
    print("Passed:", int((data["Result"] == "Pass").sum()))
    print("Failed:", int((data["Result"] == "Fail").sum()))
    print(f"Class average: {data['Average_Marks'].mean():.2f}")
    print(f"Highest average: {data['Average_Marks'].max():.2f}")
    print(f"Lowest average: {data['Average_Marks'].min():.2f}")

    top = data.loc[data["Average_Marks"].idxmax()]
    low = data.loc[data["Average_Marks"].idxmin()]
    print(f"Top performer: {top['Name']} ({top['Average_Marks']:.2f})")
    print(f"Lowest performer: {low['Name']} ({low['Average_Marks']:.2f})")

    print("\nSUBJECT-WISE AVERAGE")
    subject_averages = data[SUBJECTS].mean().sort_values(ascending=False)
    for subject, average in subject_averages.items():
        print(f"{subject}: {average:.2f}")

    print("\nTOP 5 STUDENTS")
    print(data.nlargest(5, "Average_Marks")[
        ["Name", "Average_Marks", "Grade"]
    ].to_string(index=False))

    data.to_csv("student_performance_report.csv", index=False)
    print("\nFull report saved as student_performance_report.csv")

def main():
    try:
        data = load_data(DATA_FILE)
        processed = process_data(data)
        show_analysis(processed)
    except FileNotFoundError:
        print("Error: Keep students.csv in the same folder as main.py.")
    except (ValueError, KeyError) as error:
        print("Dataset error:", error)

if __name__ == "__main__":
    main()
```

APPENDIX B: STUDENT DATA — students.csv

The following table contains the 20 fictional records used as input by the program. The original CSV file uses the same column headings shown below.

Student_ID	Name	Department	Python	Mathematics	Data_Analysis	Attendance_Percent
101	Aarav Sharma	Arts	78	82	74	91
102	Diya Verma	Commerce	88	91	85	96
103	Kabir Singh	Science	56	62	58	79
104	Ananya Patel	Arts	92	89	94	98
105	Ishaan Gupta	Commerce	43	51	48	72
106	Meera Joshi	Science	67	73	69	88
107	Arjun Rao	Arts	35	42	39	65
108	Sara Khan	Commerce	81	77	84	93
109	Vivaan Das	Science	74	70	76	81
110	Ira Mishra	Arts	95	96	92	99
111	Reyansh Jain	Commerce	61	58	64	76
112	Aditi Roy	Science	49	55	52	83
113	Krish Nair	Arts	86	80	88	90
114	Myra Kapoor	Commerce	72	75	71	87
115	Advait Sen	Science	39	33	45	61
116	Kiara Mehta	Arts	68	66	72	84
117	Rohan Yadav	Commerce	90	87	91	95
118	Saanvi Bose	Science	58	64	61	78
119	Dev Malhotra	Arts	47	44	50	70
120	Pari Chawla	Commerce	77	83	79	92

CSV header: Student_ID, Name, Department, Python, Mathematics, Data_Analysis, Attendance_Percent

Note: This appendix displays the data for readability. If running the program, the file must be saved in CSV format rather than copied from a PDF.

APPENDIX C: PROJECT DOCUMENTATION

The following is the concise project documentation prepared for the submission.

1. Project Title

Student Performance Analytics System

Student Name: Diksha Ahirwar

Course: Python with AI

Project Type: Mini Project

2. Objective

The objective of this project is to develop a Python-based system that analyzes students' marks and attendance. It calculates total marks, average marks, grades, pass/fail results, class statistics, subject-wise averages and top performers.

3. Technologies Used

- Python 3
- Pandas
- NumPy
- CSV dataset

4. Dataset Description

The dataset contains 20 fictional student records. Its columns are Student ID, Name, Department, marks in Python, Mathematics and Data Analysis, and Attendance Percentage.

5. Implementation

1. Pandas reads `students.csv` into a DataFrame.
2. NumPy calculates total and average marks.
3. The `assign_grade()` function assigns grades.
4. A loop processes students one by one.
5. Conditions determine pass/fail and attendance status.
6. Pandas calculates class statistics, subject averages and the top five students.
7. The processed report is saved to `student_performance_report.csv`.

6. Key Features

- Student-wise total and average
- Grade calculation
- Pass/fail status
- Attendance status
- Highest and lowest average
- Overall class average
- Subject-wise performance analysis
- Top five students
- CSV report export

7. Output / Screenshots

After running `main.py`, take screenshots of:

- Student-wise report
- Class summary
- Subject-wise average
- Top five students
- Saved report confirmation

Add your actual screenshots here before final submission.

8. Final Outcome

The project converts raw student records into a structured performance report. It demonstrates Python fundamentals, functions, loops, conditions, NumPy calculations and Pandas data analysis.

9. Challenges & Learning

This project provides practice in reading CSV files, matching column names, using loops and conditions, defining reusable functions, calculating statistics and saving reports. Edit this section to accurately describe any challenges you personally faced and how you solved them.

10. GitHub Repository

Repository URL: Add your actual GitHub URL here after uploading the files.

APPENDIX D: README FILE

Project title: Student Performance Analytics System

Project overview

This project analyses student marks and attendance using Python, Pandas and NumPy. It reads 20 fictional sample student records from students.csv, calculates totals and averages, assigns grades, checks pass/fail status and attendance, summarises class performance, identifies top performers, and saves the processed results to student_performance_report.csv.

Features

- Reads sample student records from students.csv.
- Calculates total and average marks.
- Assigns grades using the defined grading scale.
- Passes a student only if every subject mark is at least 40.
- Checks attendance against the 75% threshold.
- Reports class average, highest/lowest average, subject averages and top five students.
- Saves a processed report as student_performance_report.csv.

Technologies

Python 3, Pandas, NumPy and CSV.

How to run the project

- Keep main.py and students.csv in the same folder.
- Install the required libraries in your Python environment: pip install pandas numpy
- Run the program: python main.py
- Review the displayed summary and the generated student_performance_report.csv file.

Grading scale

Average marks	Grade
90 and above	A+
80 to below 90	A
70 to below 80	B
60 to below 70	C
50 to below 60	D
Below 50	E

Pass rule

A student passes only if they score at least 40 in all three subjects.

Dataset note

The 20 student names and marks are fictional sample data for demonstration.

Student details

Student Name: Diksha Ahirwar
Course: Python with AI

REFERENCES AND LEARNING RESOURCES

- Python Documentation — https://docs.python.org/3/
- Pandas Documentation — https://pandas.pydata.org/docs/
- NumPy Documentation — https://numpy.org/doc/
- Sample dataset: students.csv, fictional records prepared for this mini project.

END OF PROJECT REPORT